

Week 6: Exploring Data

September 30, 2026

WHY EXPLORE

Exploratory analysis *forms* hypotheses and improves the quality of your **questions** — that’s this class. **Confirmatory** analysis *tests* hypotheses and improves the quality of your **answers** — that’s a stats class. EDA is iterative: summarize, visualize, ask a sharper question, repeat.

“An *approximate* answer to the *right* question is better than an *exact* answer to the *wrong* question.” — John Tukey

A number alone means nothing. “24,901” becomes information only with its labels: *Earth’s circumference at the equator, in miles*.

TYPES OF DATA

The type decides which summaries and charts make sense.

Type	Kind	Key property	Example
Nominal	Categorical	Order doesn’t matter	<code>country</code>
Ordinal	Categorical	Order matters; gaps unequal	Race placement
Interval	Numerical	No true zero; can’t say “double”	Temperature
Ratio	Numerical	True zero; “double” makes sense	Height, speed

Two questions sort any variable. **Categorical:** does the order matter? (yes = ordinal). **Numerical:** is there a baseline zero? (yes = ratio).

Watch the encoding

- Numbers that are categories:** dummy codes (1 / 0), or regions coded 1–4. Averaging these is meaningless.
- Counts:** ratio data measured in whole units (eggs, heartbeats, age).
- Time** can be ordinal (Jan., Feb.), interval (Jan. 1, Jan. 2), or ratio (30 sec, 60 sec).

CENTRALITY & VARIABILITY

Measure	What it is	Robust?
Mean	Sum of values / number of values	No
Median	Middle value; 50% above and below	Yes
Standard deviation	Typical distance from the mean	No
IQR	$Q_3 - Q_1$, the middle 50%	Yes
Range	Max – min	No

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

Beware the flaw of averages. Planes hit birds at a *mean* height near 1,000 ft, but most strikes happen under 100 ft — the mean sits above most of the data. And two warehouses can share a mean and median shipping time while one ranges 3–5 days and the other 1–10. The spread is the story.

Outliers fall below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$; **extreme values** use $3 \times \text{IQR}$. Change one value of `c(3, 3, 4, 5, 5, 6, 6, 7, 8, 9)` to 20: the mean jumps 5.6 to 6.7 and the SD 2.0 to 4.9, but the median stays 5.5. **Skewed or outlier-heavy data: use median and IQR.**

```
wildlife_impacts %>%
  summarise(mean = mean(height, na.rm = TRUE),
            median = median(height, na.rm = TRUE),
            sd = sd(height, na.rm = TRUE),
            iqr = IQR(height, na.rm = TRUE))
summary(wildlife_impacts$height) # quick view
```

THE TYPE SETS THE SUMMARY

Type	Measures	Chart
Nominal	Counts, proportions	Bars, sorted
Ordinal	Counts; median, mode, IQR	Bars, in order
Continuous	Mean, median, SD, IQR, range	Histogram, boxplot

```
wildlife_impacts %>%
  count(operator, sort = TRUE) %>%
  mutate(p = n / sum(n)) %>%
  ggplot() +
  geom_col(aes(x = n, y = reorder(operator, n)))
```

Histograms show skewness and the number of modes; try a few **bins** values. **Boxplots** show outliers and compare groups. Use both — each hides what the other shows.

ALWAYS PLOT THE DATA

Anscombe’s quartet and the datasaurus share means, SDs, and correlations to two decimals, yet look nothing alike once plotted. Summary numbers compress; charts reveal shape, clusters, and outliers.

A summary statistic is a claim about the data — a chart is how you check it. Before you trust a mean or an r (yours or an agent’s), look at the histogram, boxplot, or scatterplot behind it.

CORRELATION

Pearson’s r is the covariance of x and y divided by the product of their standard deviations, so $-1 \leq r \leq 1$. The closer to ± 1 , the stronger the relationship.

$ r $	Strength
0.1–0.3	Weak
0.3–0.5	Moderate
0.5–0.8	Strong
0.8–1.0	Very strong

	Pearson	Spearman
Data	Interval or ratio	Ordinal, interval, or ratio
Detects	Linear relationship	Monotonic relationship
Outliers	Highly sensitive	Robust (uses ranks)

```
cor(x, y, method = "pearson", use = "complete.obs")
cor(x, y, method = "spearman", use = "complete.obs")
```

Report both r and r^2 . r^2 is the share of variance in one variable explained by the other. Parent and child height: $r = 0.32$, so $r^2 \approx 0.10$ — 10% of the variance explained.

Where correlation misleads

- Correlation isn’t causation.** X may cause Y, Y may cause X, or a third variable Z may cause both.
- Simpson’s paradox.** Penguin body mass and bill depth correlate *negatively* overall but *positively* within each species. Color by group.
- Dual axes** can manufacture spurious correlations. Use one axis.
- History.** Correlation came from Galton and Pearson’s work in eugenics.

VISUALIZING RELATIONSHIPS

Variables	Summary	Chart
2 categorical	<code>count(a, b)</code> , then <code>pivot_wider()</code>	Dodged bars, <code>fill = 2nd var</code>
2 continuous	<code>cor()</code>	Scatterplot
1 continuous, 1 categorical	Grouped <code>summarise()</code>	Boxplot

```
# Label a scatterplot with its correlation
r <- cor(mtcars$mpg, mtcars$hp)
scatterplot +
  annotate(geom = 'text', x = 25, y = 310, hjust = 0,
    label = paste('r = ', round(r, 2)))
```

Scan many variables at once with GGally:

```
mtcars %>% ggcorr(label = TRUE, label_round = 2,
  method = c("pairwise", "spearman"))
mtcars %>%
  select(mpg, cyl, disp, hp, wt) %>%
  ggpairs()
```

Tip: `theme_set(theme_bw(base_size = 20))` at the top of a file sets the theme for every plot after it.

AN EDA LOOP FOR A NEW DATA SET

1. `glimpse()` it. Note each variable's type and what it measures.
2. Summarize each variable: counts for categories; median, IQR, and range for numbers.
3. Plot each variable: bars, a histogram, or a boxplot.
4. Write a question about two variables, then pick the chart that answers it.
5. Let what you see raise the next question. Repeat.

DIRECTING AN AGENT TO EXPLORE

An agent will happily report a mean and an r . Your job is to ask for the evidence behind them and to choose the robust measure when the data call for it:

- “Summarize `height`: mean, median, SD, and IQR. Show a histogram and a boxplot, and tell me whether it's skewed.”
- “Compute Pearson and Spearman correlations for every numeric pair. Flag pairs where they disagree, and plot those.”
- “Plot `body_mass_g` against `bill_depth_mm`, colored by species. Does the trend hold within each group?”

ERRORS YOU MAY HIT THIS WEEK

Message / symptom	Usual cause
<code>mean()</code> or <code>sd()</code> returns NA	Missing values; add <code>na.rm = TRUE</code>
<code>cor()</code> returns NA	Missing values; add <code>use = "complete.obs"</code>
“argument is not numeric or logical”	Column is <code><chr></code> ; check <code>glimpse()</code>
“Removed N rows containing non-finite values”	NAs dropped from the plot; count them
“ <code>stat_bin()</code> using <code>bins = 30</code> ”	Default bins; try a few bins values
Month drawn as a continuous axis	Numbers as categories; wrap in <code>as.factor()</code>
Bars in alphabetical order	Sort with <code>reorder(operator, n)</code>
<code>ggcorr()</code> : “not numeric and were ignored”	Text columns skipped; expected

BEFORE NEXT CLASS

- **HW5: Visualizing Information**, due **Oct 5**.
- **MP1: Clean & Verify**, due **Oct 4** (9% of grade).
- **Quiz 3** on Oct 14, covering Weeks 6–7.

You should be able to:

- Classify a variable as nominal, ordinal, interval, or ratio.
- Pick mean/SD or median/IQR, and explain why.
- Choose a bar chart, histogram, or boxplot for one variable.
- Compute and interpret Pearson and Spearman correlations, plus r^2 .
- Choose the right chart for a two-variable relationship.